

Improving the SegFormer Decoder for Semantic Segmentation by Adding a Depthwise Convolution Block

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Abstract

Efficient semantic segmentation networks are usually improved by modifying the token mixer of the encoder. In a controlled study on Oxford-IIIT Pet, four token mixers are substituted into an identical SegFormer-B0 architecture under a single training protocol. Mixers that inherit pretrained attention weights reach the same accuracy (border-class IoU within 0.0003) while differing in throughput by a factor of six; the token mixer therefore determines the cost of the model but not its accuracy. A per-class error analysis locates the accuracy bottleneck in the decoder. The all-MLP decoder of SegFormer consists of 1×1 convolutions only, so its receptive field on the fused feature map equals one position. Inserting a single depthwise 3×3 convolution after the fusion step adds 2816 parameters (0.08% of the model) without measurable latency and raises mIoU from 0.8095 to 0.8161, border IoU from 0.6004 to 0.6137 and mask mAP_{50:95} from 0.8198 to 0.8270. A paired ablation isolates the effect of this convolution. The modified model is also compared with DeepLabV3+ and YOLOv8s-seg under a shared binary foreground protocol.

Keywords: semantic segmentation, vision transformer, SegFormer, decoder design, receptive field, depthwise convolution

1. Introduction

Semantic segmentation assigns a class label $\hat{y}_{ij} = \arg \max_c z_{cij}$ to every pixel of an image $x \in \mathbb{R}^{3 \times H \times W}$, where $z \in \mathbb{R}^{C \times H \times W}$ are the predicted logits. The output has the spatial extent of the input, so a segmentation model must combine global semantic context with precise localisation.

SegFormer [1] combines a hierarchical transformer encoder with a minimal all-MLP decoder and provides a strong accuracy–efficiency trade-off. Subsequent work has mainly replaced the encoder token mixer with state space scans [2], linear attention [3] or convolutional attention [4], which assumes that the token mixer determines segmentation accuracy. The present work tests this assumption under controlled conditions and, on the basis of the result, examines the decoder instead.

The contributions are as follows:

1. a controlled comparison showing that the choice of token mixer does not affect accuracy when pretrained weights are inherited;
2. a receptive-field analysis identifying the all-MLP decoder as the accuracy bottleneck;
3. a decoder modification with one depthwise 3×3 convolution (hereafter SegFormer-B0-DW) that improves all quality metrics at a cost of 2816 parameters.

2. Background

SegFormer encoder. The MiT encoder [1, 5, 6] processes overlapping patches in four stages with strides 4, 8, 16, 32 and, for the B0 configuration, channel widths 32, 64, 160, 256. Self-attention over N tokens costs $O(N^2d)$.

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SegFormer reduces keys and values spatially by a ratio R ,

$$X' = \text{SR}(X, R) \in \mathbb{R}^{(N/R^2) \times d}, \quad K = X' W_K, \quad V = X' W_V,$$

which lowers the cost to $O(N^2 d / R^2)$ with $R = 8, 4, 2, 1$ across the stages.

All-MLP decoder. Let $F^{(k)} \in \mathbb{R}^{C_k \times H_k \times W_k}$ be the output of stage k . The decoder projects each stage to a common width D , resizes it to stride four and fuses the result:

$$U^{(k)} = \text{Resize}(W_p^{(k)} *_{1 \times 1} F^{(k)}), \quad G = W_f *_{1 \times 1} [U^{(1)} \parallel U^{(2)} \parallel U^{(3)} \parallel U^{(4)}].$$

Batch normalisation, ReLU, a 1×1 classifier and bilinear upsampling by four follow. All convolutions in this path have 1×1 kernels.

Reference models. DeepLabV3+ [7] uses a ResNet-50 backbone [8] with atrous spatial pyramid pooling, in which a 3×3 kernel with dilation r covers $2r + 1$ positions. YOLOv8-seg [9] is an instance segmentation model that predicts masks $m_n = \sigma(\sum_k \alpha_{n,k} P_k) \odot \mathbb{I}[b_n]$ from shared prototypes P_k cropped by a bounding box b_n . The instance formulation cannot represent classes without object instances, such as the boundary ring used in this work; the comparison with YOLOv8s-seg is therefore performed on a binary foreground task.

3. Methods

Dataset and protocol. Experiments use Oxford-IIIT Pet [10] with three classes: background, pet and border (the ambiguity ring around each animal). The official split is used: 3680 trainval images (10% held out for monitoring) and 3669 test images, resized to 256×256 . All models are trained with AdamW [11], learning rate 3×10^{-4} (multiplied by 0.1 for pretrained encoder parameters), weight decay 0.01, polynomial decay with 5% warmup, gradient clipping at 1.0, batch size 16, 20 epochs, flip, scaled crop and brightness augmentation, pixel-wise cross entropy loss and a fixed seed 1337.

Metrics. With per-class pixel counts TP_c , FP_c , FN_c accumulated over the test set,

$$\text{IoU}_c = \frac{\text{TP}_c}{\text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad \text{Dice}_c = \frac{2 \text{TP}_c}{2 \text{TP}_c + \text{FP}_c + \text{FN}_c}, \quad F_{\beta,c} = \frac{(1 + \beta^2) P_c R_c}{\beta^2 P_c + R_c},$$

where P_c and R_c are precision and recall; $\beta = 2$ is reported. Mask $\text{mAP}_{50:95}$ is computed over connected foreground components as the mean of 101-point interpolated AP at IoU thresholds 0.50, 0.55, ..., 0.95. Latency is the mean of 25 forward passes at batch size one after 8 warmup passes.

4. Effect of the Token Mixer

The original efficient self-attention [1] is compared with four token mixers substituted into an identical architecture: reduced-key linear attention [3], fixed-budget gated attention with keys and values pooled to an 8×8 grid, convolutional attention [4] and a selective state space scan [2]. Table 1 shows that mixers inheriting the pretrained projection weights reach border IoU between 0.6004 and 0.6006. Mixers with a different parameterisation start from random initialisation and lose accuracy. Training the same architecture with and without inherited weights gives a difference of +0.1018 mIoU for linear attention and +0.0430 for fixed-budget attention, which exceeds the spread between any two mixers. The token mixer thus determines inference cost but not accuracy.

5. Proposed Decoder Modification

5.1. Error analysis

The per-class IoU of the baseline is 0.9412 for background, 0.8869 for pet and 0.6004 for border. Two explanations of the border deficit are excluded. *Decoder resolution:* passing the ground truth through the same stride-four bottleneck (one-hot encoding, bilinear downsampling and upsampling, arg max) gives an upper bound of 0.9101 border IoU, far above the model value. *Calibration:* scaling the border logit by 2.5 raises border IoU by 0.0095 but lowers mIoU from 0.7989 to 0.7956; the mean predicted border probability on border pixels is already 0.6496. The error is therefore caused by inaccurate localisation of the ring rather than by resolution or calibration.

Table 1: Token mixers in an identical architecture under one training protocol.

Token mixer	Inherits	mIoU	Border IoU	mAP _{50:95}	FPS
Efficient self-attention	yes	0.8095	0.6004	0.8198	156.9
Reduced-key linear attention	yes	0.8093	0.6006	0.8190	136.1
Fixed-budget gated attention	yes	0.8092	0.6005	0.8186	143.9
Convolutional attention	no	0.7871	0.5651	0.7826	141.2
Selective state space scan	no	0.7549	0.5202	0.7345	24.0

SEGFORMER-B0 WITH A DEPTHWISE 3X3 CONVOLUTION IN THE DECODER

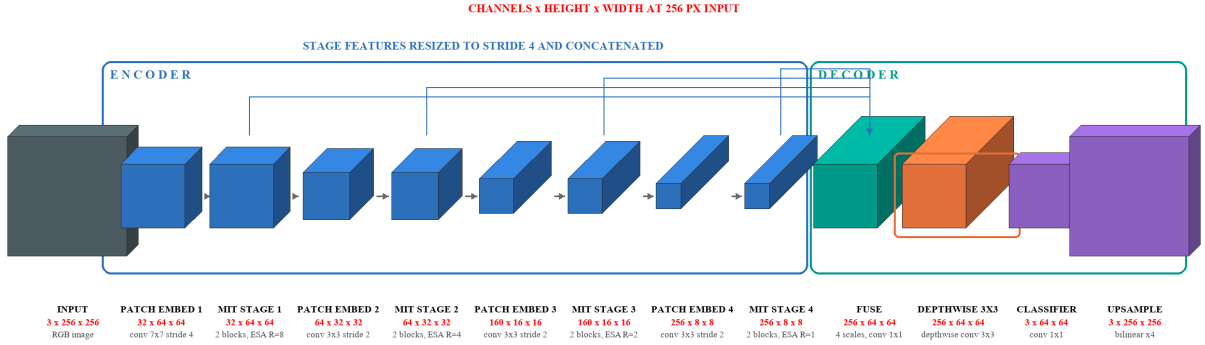


Figure 1: SegFormer-B0-DW. The MiT-B0 encoder and the all-MLP fusion are unchanged; a depthwise 3×3 convolution is inserted between the fusion and the classifier.

5.2. Receptive field of the decoder

Let $G \in \mathbb{R}^{D \times H/4 \times W/4}$ be the fused feature map and ρ the receptive field of the decoder on G . Since all operations after fusion are 1×1 convolutions, normalisation or pointwise nonlinearities,

$$z_{:,i,j} = g(G_{:,i,j}), \quad \rho = 1,$$

and each stride-four cell is classified independently of its neighbours. Determining whether a cell lies on the boundary requires information about adjacent cells. A depthwise 3×3 convolution [12] inserted after fusion gives

$$z_{:,i,j} = g\left(\sum_{|a| \leq 1} \sum_{|b| \leq 1} w_{a,b} \odot G_{:,i+a,j+b}\right), \quad \rho = 3,$$

which corresponds to 12 pixels at input resolution. The block contains $D \times 3 \times 3 = 2304$ weights and 512 normalisation parameters, 2816 in total, against 3 714 915 parameters of the baseline. The resulting architecture is shown in Fig. 1.

6. Results

6.1. Paired ablation

Two models differing only by the depthwise convolution are trained with identical widths, protocol and seed (Table 2). The largest gain is observed for the border class (+0.0121 IoU), while background and pet improve by 0.0021 and 0.0046, in agreement with the receptive-field analysis: region interiors are classified correctly from a single feature vector, whereas boundaries require neighbouring context.

Table 2: Paired ablation: the two decoders differ by one depthwise 3×3 convolution.

Decoder	Params	mIoU	Border IoU	mAP _{50:95}	FPS
1×1 only, $\rho = 1$	3 714 915	0.8102	0.6016	0.8205	163.8
with depthwise 3×3 , $\rho = 3$	3 717 731	0.8161	0.6137	0.8270	161.9
Difference	+2816	+0.0059	+0.0121	+0.0065	-1.9

Table 3: Comparison on the Oxford-IIIT Pet test split (3669 images). FG denotes the binary foreground task.

Model	Params	FG IoU	FG Dice	FG F2	mAP _{50:95}	mIoU	Border IoU	FPS
SegFormer-B0-DW	3.72 M	0.9227	0.9598	0.9569	0.8270	0.8161	0.6137	161.9
SegFormer-B0-BR	3.50 M	0.9225	0.9597	0.9560	0.8271	0.8162	0.6138	116.0
SegFormer-B0	3.71 M	0.9199	0.9583	0.9558	0.8198	0.8095	0.6004	156.9
YOLOv8s-seg	11.82 M	0.8397	0.9129	0.8801	0.6196	—	—	152.2
DeepLabV3+-R50	42.00 M	0.8329	0.9088	0.8724	0.6398	—	—	27.4

6.2. Comparison with the baseline and reference models

DeepLabV3+ and YOLOv8s-seg have no border class, so all models are additionally evaluated on a binary foreground task in which pet and border are merged; VOC and COCO cat and dog classes are mapped to foreground for the reference models. Binary scores of the SegFormer models are derived from their three-class confusion matrices (deviation from a direct measurement below 0.0003). Table 3 and Fig. 2 summarise the results.

SegFormer-B0-DW exceeds the baseline in all quality metrics; its latency coincides with that of the baseline within measurement noise at all resolutions (Fig. 3). The reference models were evaluated as published checkpoints without fine-tuning on this dataset, so the gap of about 0.08 in foreground IoU reflects in-domain training rather than architectural inferiority. DeepLabV3+ requires 11 times more parameters and is six times slower than SegFormer-B0-DW. YOLOv8s-seg has the lowest mask AP (0.6196) because a missed detection removes the entire object from the prediction.

7. Discussion

Mechanism. SegFormer-B0-BR in Table 3 denotes an alternative decoder with a full-resolution detail branch and a supervised boundary gate, $z = z_{\text{sem}} + \sigma(b) \odot \Delta(\text{detail}, \text{semantic})$. Removing the boundary gate and its supervision leaves mIoU unchanged (0.8167 against 0.8162), as does removing the detail branch (0.8163). Both variants retain a depthwise convolution in the fusion path, which the paired ablation identifies as the source of the gain. The improvement is therefore explained by the enlarged receptive field of the decoder rather than by the additional branches, which reduce throughput from 161.9 to 116.0 FPS.

Limitations. The evaluation uses one dataset with three classes and one salient object per image; the results may not transfer to ADE20K or Cityscapes. The baseline is a fine-tuned SegFormer-B0 rather than a state-of-the-art model. All timings were obtained on Apple Silicon with the Metal backend and may differ on other hardware. The reference models were not fine-tuned on the dataset.

8. Conclusion

Under a fixed training protocol, token mixers that inherit pretrained weights yield the same segmentation accuracy while differing by a factor of six in throughput. The accuracy bottleneck of SegFormer lies in its all-MLP decoder, whose receptive field on the fused feature map equals one position. Adding a single depthwise 3×3 convolution after the fusion step increases the receptive field to three positions and raises mIoU from 0.8095 to 0.8161, border IoU from 0.6004 to 0.6137 and mask mAP_{50:95} from 0.8198 to 0.8270 at a cost of 2816 parameters and no measurable latency.

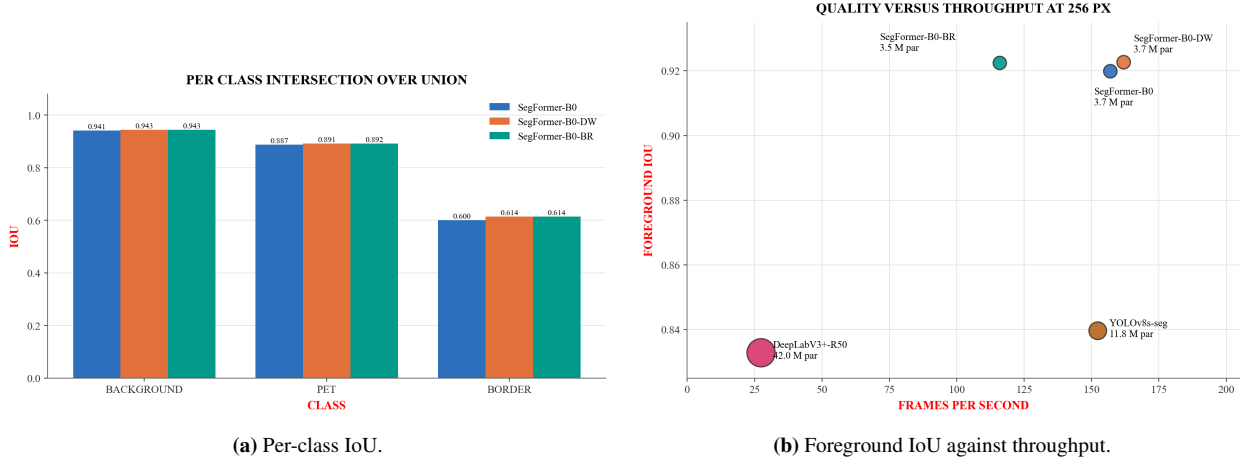


Figure 2: Per-class quality and the quality–throughput trade-off of the compared models.

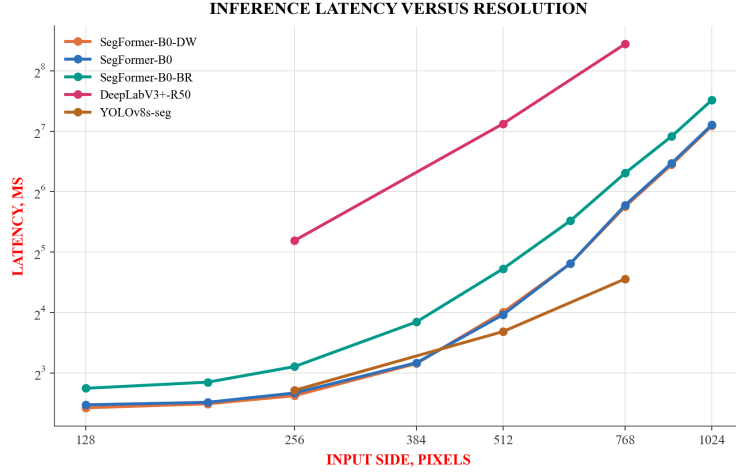


Figure 3: Inference latency against input resolution.

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